



Technical Note

Analysis of transient conduction and radiation problems using lattice Boltzmann and finite volume methods

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ABSTRACT

In this paper, the lattice Boltzmann method (LBM) and the finite volume method (FVM) are coupled to solve transient conduction and radiation heat transfer problems. They are used to solve the energy equation and get the radiative heat source, respectively. To avoid additional interpolations between these two solvers when adding the radiative heat source to the energy equation, we propose to use the same grid systems for the LBM and FVM. This is achieved by adopting a half way bounce back boundary scheme for LBM. Good agreement with available literature is obtained for both 1D and 2D conditions. Effects of conduction–radiation parameter and scattering albedo on the transient temperature distribution are presented and discussed. The method proposed by this study is efficiently able to solve combined conduction and radiation problems.

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1. Introduction

The lattice Boltzmann method (LBM) is becoming a useful and promising computational tool and has found many applications in fluid flow and heat transfer analyses. LBM is based on the mesoscopic kinetic equations and thus has many attractive advantages over conventional computational tools. These include simple algorithm procedures, efficient parallel computation and easy treatment of irregular geometries and boundary conditions.

LBM has gained momentum in heat transfer problems [1–13] and been extended to solve these problems involving thermal radiation, mainly done by Mishra and co-workers [14–16]. Mishra solved a 1D planar transient conduction and radiation heat transfer problems by using the LBM (for energy equation) and the discrete transfer method (for radiative transfer equation (RTE)). Other RTE solvers, such as the collapsed dimension method [16], finite volume method (FVM) [14] and discrete ordinates method (DOM) [15] were also combined with the LBM to solve the coupled heat transfer problems. They have applied the LBM to a broad range of problems and the reader is referred to them for detail. This paper focuses on combined conduction and radiation problems.

Radiation simulation usually consists of two aspects, namely, the direction nature and the spectral nature. This paper focuses on the solution of the RTE, so the spectral aspect is neglected. One

is referred to [17,18] on this issue for detail. The RTE is mathematically an integro-differential equation and is difficult to resolve even without the consideration of the spectral variations of the radiative properties. There exist only a few analytical solutions for some simple problems and thus numerical methods are usually adopted to solve the RTE. The DOM and the FVM are the most popular ones due to their balance between numerical accuracy and computational resources. In fact, they share many features and chiefly differ on the angular discretization procedure [19]. The FVM is more popular due to its compatibility with the FVM grids used in the solution of momentum and energy equations. The FVM or other RTE solvers coupled with CFD solvers have been used to solve combined radiation and conduction/convection problems. Here we will review some of the recent publications. Mishra and Roy [14] solved transient conduction and radiation heat transfer problems by using the LBM and FVM, which were used to solve the energy equation and RTE, respectively, while Mondal and Mishra [15] used DOM for getting radiative information instead of FVM. Aouled-Dlala et al. [20] developed a new technique to treat the angular derivative term when applied to spherical and cylindrical media by using DOM and the boundary value problem with a finite difference algorithm. Chaabane et al. [21] used the LBM and the control volume finite element method (CVFEM) to solve transient conduction and radiation heat transfer in a cylindrical medium. They were used to solve the energy equation and RTE, respectively. Sun et al. [22] used the Chebyshev collocation spectral method (CSM) to solve three-dimensional transient combined conduction and radiation heat transfer, and the CSM was used for spatial discretization of RTE and energy equation and DOM for angular

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discretization of RTE. Zhang et al. [23,24] developed a natural element method for radiative heat transfer in irregular geometries for 2D and 3D enclosures. Maroufi and Aghanajafi [25] analysed the solidification of a 2D semitransparent material by using the LBM for both conduction and radiation transfer. Luo et al. [26] applied the LBM and collocation meshless methods to solve laminar mixed convection in an eccentric annulus with participating media. The LBM was used to compute the density, velocity, and thermal fields, while the DCM provided the radiation information. This hybrid approach is able to treat irregular geometries.

As seen from above literature survey, since combination of the LBM and FVM for conduction–radiation heat transfer problems by Mishra and Roy [14], LBM has been applied by many other researchers. But as pointed out by Asinari et al. [27], the computational grids of the LBM and the FVM in [14] were not overlapped, which means the radiative heat source obtained by the FVM had to be interpolated to the LBM solver. The additional interpolation operations need more computational steps and break the simplicity of the LBM. To address this problem, we propose in this work to use the same grid systems for the LBM and FVM by using a relatively new boundary treatment for the thermal LBM, in which we superpose the grids of these two solvers and thus drop out the interpolation operations. The purpose of this study is to apply this method to solve the combined conduction and radiation problems in 1D planar and 2D square enclosure. Extensions to combined convection and radiation problems are not difficult.

The remainder of this paper reviews the mathematical formulations of the LBM and the FVM, and presents the coupling implementations of these two methods. The boundary treatment of LBM used in this paper is described in detail. The validation of the obtained results is discussed. Conclusions are then drawn.

2. Mathematical model

The transient heat conduction equation in the presence of radiation is:

$$\rho c_p \frac{\partial T}{\partial t} = k \nabla^2 T - \nabla \cdot \mathbf{q}_r \quad (1)$$

where ρ , c_p and k are the density, the specific heat and the thermal conductivity of the medium, respectively. They are assumed to be constant in this work. $\mathbf{q}_r$ is the radiative heat flux and can be obtained from the FVM. In this section, we give the formulation of the LBM and the FVM and the proposed coupling strategy between them.

2.1. Lattice Boltzmann method (LBM)

For the LBM to simulate the heat transfer process, one distribution function is enough. Its evolution is governed by the lattice Boltzmann equations with the single relaxation time approximation for the collision operator,

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) - f_\alpha(\mathbf{x}, t) = -\frac{1}{\tau} [f_\alpha(\mathbf{x}, t) - f_\alpha^{eq}(\mathbf{x}, t)] \quad (2)$$

where $f_\alpha(\mathbf{x}, t)$ is the population at location $\mathbf{x}$ and time t and the subscript α denotes the lattice direction.

For conduction problems, D1Q2 or Q1Q3 are usually used for one dimensional (1D) and D2Q9 for two dimensional (2D) problems. The equilibrium distributions $f_\alpha^{eq}(\mathbf{x}, t)$ in α direction is $f_\alpha^{eq}(\mathbf{x}, t) = w_\alpha T$, and w_α is the weight coefficient in this direction.

The macroscopic temperature can be readily obtained from

$$T(\mathbf{x}, t) = \sum_\alpha f_\alpha(\mathbf{x}, t) \quad (3)$$

To take into account the volumetric radiation, a radiative source term is added to the evolution equation

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta t, t + \delta t) = f_\alpha(\mathbf{x}, t) - \frac{1}{\tau} [f_\alpha(\mathbf{x}, t) - f_\alpha^{eq}(\mathbf{x}, t)] - \left(\frac{\delta t}{\rho c_p} \right) w_\alpha \nabla \cdot \mathbf{q}_r \quad (4)$$

The diffusion equation, Eq. (1) with radiative heat source can be obtained from the above equations using the Chapman–Enskog analysis.

2.2. Finite volume method (FVM)

The radiative transfer equation (RTE) in any direction $\hat{s}$ is:

$$\frac{dI}{ds} = -(\kappa_a + \sigma_s)I + \kappa_a \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_{\Omega'=4\pi} I\Phi(\Omega, \Omega') d\Omega' \quad (5)$$

where κ_a is the absorption coefficient, σ_s is the scattering coefficient and Φ is the scattering phase function. It can be rewritten in the following form:

$$\frac{dI}{ds} = -\beta I + S \quad (6)$$

where $\beta = \kappa_a + \sigma_s$ is the extinction coefficient and S is the source term given by

$$S = \kappa_a \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_{\Omega'=4\pi} I\Phi(\Omega, \Omega') d\Omega' \quad (7)$$

Integrating Eq. (5) along the solid angle, we get

$$\frac{\partial I^m}{\partial x} D_x^m + \frac{\partial I^m}{\partial y} D_y^m + \frac{\partial I^m}{\partial z} D_z^m = -\beta I^m \Delta \Omega^m + S^m \Delta \Omega^m \quad (8)$$

where D^m is given by

$$D^m = \int_{\Delta \Omega^m} (\hat{n} \cdot \hat{s}^m) d\Omega \quad (9)$$

where $\hat{n}$ is the unit vector of the (i, j, k) basis in Cartesian coordinates. Eq. (8) can be solved with finite volume method.

The radiative heat source $\nabla \cdot \mathbf{q}_r$ in the energy equation is:

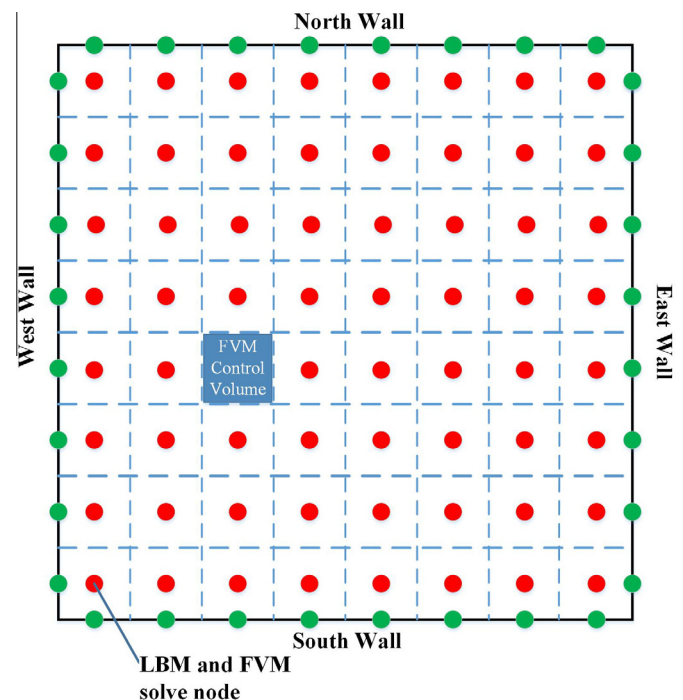


Fig. 1. 2D LBM and FVM control volume.

Table 1

Dimensionless temperature at dimensionless time $\zeta = 0.05$ for $\tau_L = 1.0$, $\omega = 0.5$, $T_E = 0.0$ and $N = 0.1$ for two sets of wall emissivities.

ε_W	ε_E	Investigators	Temperature		
			$X = 0.25$	$X = 0.5$	$X = 0.75$
1.0	1.0	Sutton [30]	0.4888	0.1778	0.0591
		Tsai and Lin [31]	0.4889	0.1773	0.0588
		Talukdar and Mishra [16]	0.4892	0.1768	0.0585
		Mishra and Roy [14]	0.4897	0.1771	0.0587
		Mondal and Mishra [15]	0.4898	0.1769	0.0583
		LBM–FVM (Present)	0.4894	0.1771	0.0585
1.0	0.0	Sutton [30]	0.5030	0.2005	0.0833
		Tsai and Lin [31]	0.5031	0.2001	0.0830
		Talukdar and Mishra [16]	0.5033	0.1995	0.0824
		Mishra and Roy [14]	0.4996	0.1991	0.0820
		Mondal and Mishra [15]	0.5025	0.1958	0.0800
		LBM–DOM (Present)	0.5028	0.1992	0.0823

$$\nabla \cdot \mathbf{q}_r = \beta(1-\omega) \left(4\pi \frac{\sigma T^4}{\pi} - G \right) \quad (10)$$

where G is the incident radiation and can be computed as following:

$$G = \int_0^{4\pi} I d\Omega \quad (11)$$

and $\omega = \sigma_s/\beta$ is the scattering albedo. Once the intensity distribution are known, the heat flux can also be obtained:

$$\mathbf{q}_r = \int_0^{4\pi} I \hat{\mathbf{s}} d\Omega \quad (12)$$

2.3. Coupling of LBM and FVM

In this section, we describe the coupling implementations between the LBM and the FVM. In Mishra and Roy [14], the lattice grid and the FVM grid are staggered and interpolations are needed

between these two methods. Special treatments are also needed for the nodes at the boundaries and four corners. We propose to solve the radiative transfer equation and energy equation in the same grid point by superposing the LBM and the FVM grid systems, which can be realized by the half way bounce back boundary treatment in the LBM, as shown in Fig. 1. Using the half way bounce back scheme is as easy as the stand bounce back scheme.

Zhang et al. [28] developed a half way bounce back scheme for the convection system. For the Dirichlet boundary conditions, the distribution function entering from the wall is constructed as

$$f_{\alpha}(\mathbf{x}_f, t + \delta t) = -f_{\alpha}^+(\mathbf{x}_f, t) + 2\omega_{\alpha}T_{\alpha} \quad (13)$$

where $f_{\alpha}^+(\mathbf{x}_f, t)$ is the post-collision distribution of the fluid node. One is referred to [29] for the conditions with moving wall. By using this coupling method, the interpolations between the LBM and the FVM and the special treatments can be eliminated.

3. Results and discussions

In this section, we give the results of combined radiation–conduction problems by using the foregoing coupled method. We consider a 1D planar medium and 2D square medium in this paper, which are usually adopted by other investigators to test the capabilities of a numerical method. We also assume the physical properties of the medium as constant. Effects of the scattering albedo and conduction–radiation number on the temperature and heat flux are analyzed.

3.1. 1D cases

For the 1D planar medium, the system is initially at temperature T_E , and the left wall temperature rises to T_W ($>T_E$) when $t \geq 0$. Under this condition, the heat transfer will take place by conduction and radiation. The temperature of the medium keeps increasing until it reaches steady state (SS), which means the

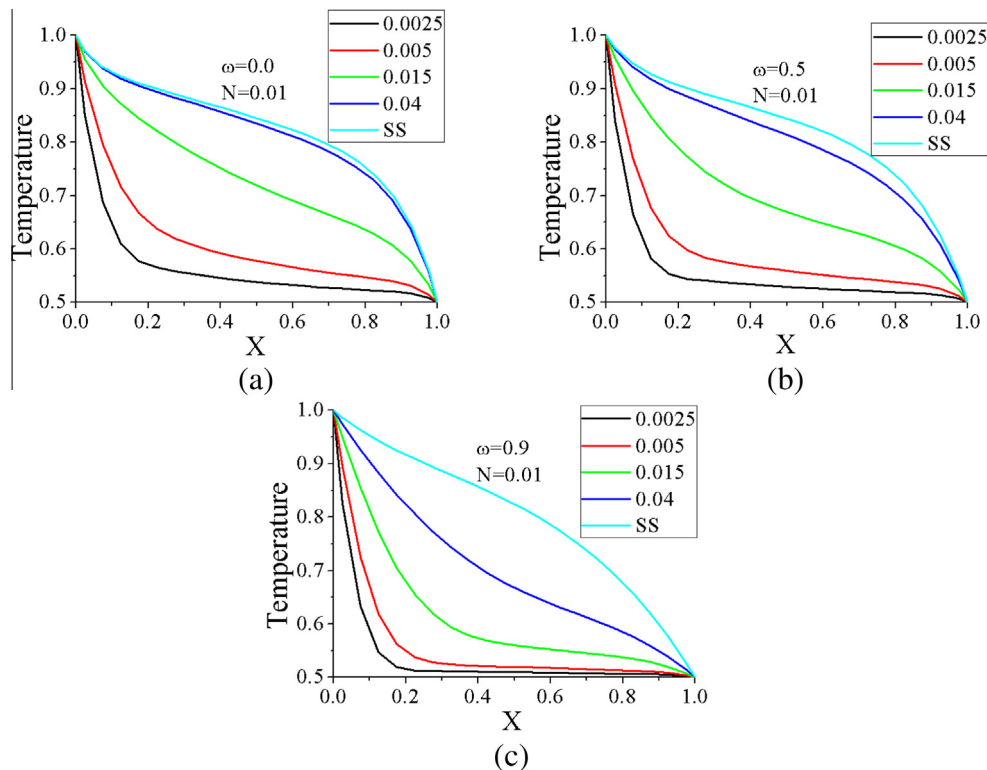


Fig. 2. Transient dimensionless temperature distribution in the planar medium for different scattering albedos (a) $\omega = 0.0$, (b) $\omega = 0.5$, and (c) $\omega = 0.9$.

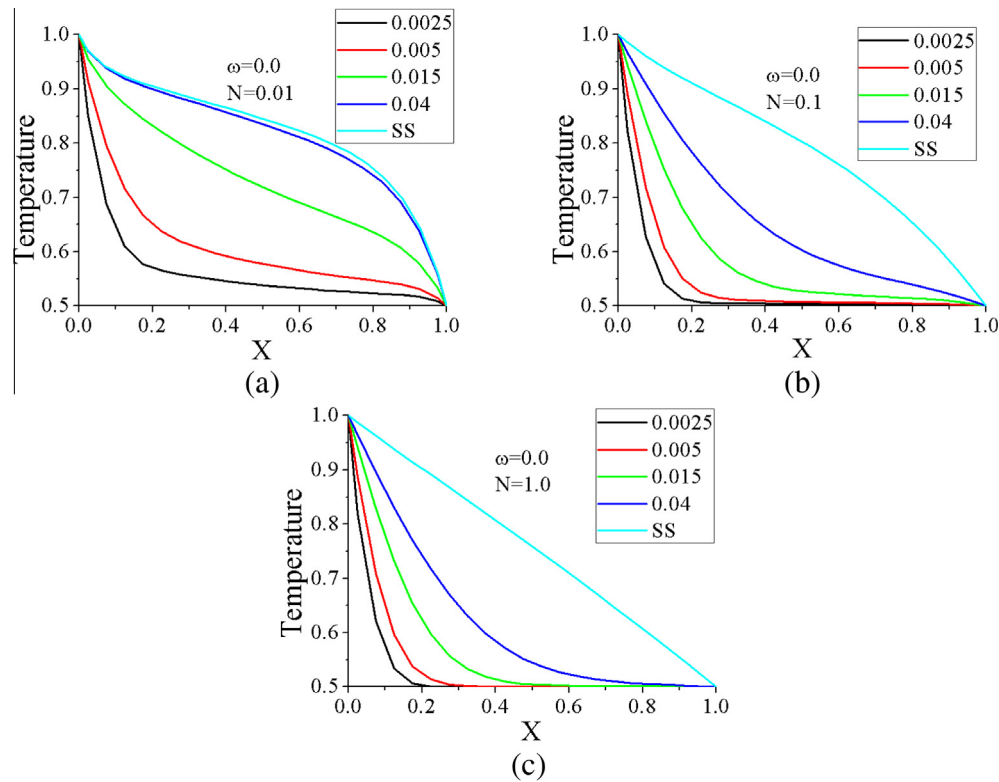


Fig. 3. Transient dimensionless temperature distribution in the planar medium for different conduction radiation parameters (a) $N = 0.01$, (b) $N = 0.1$, (c) $N = 1.0$.

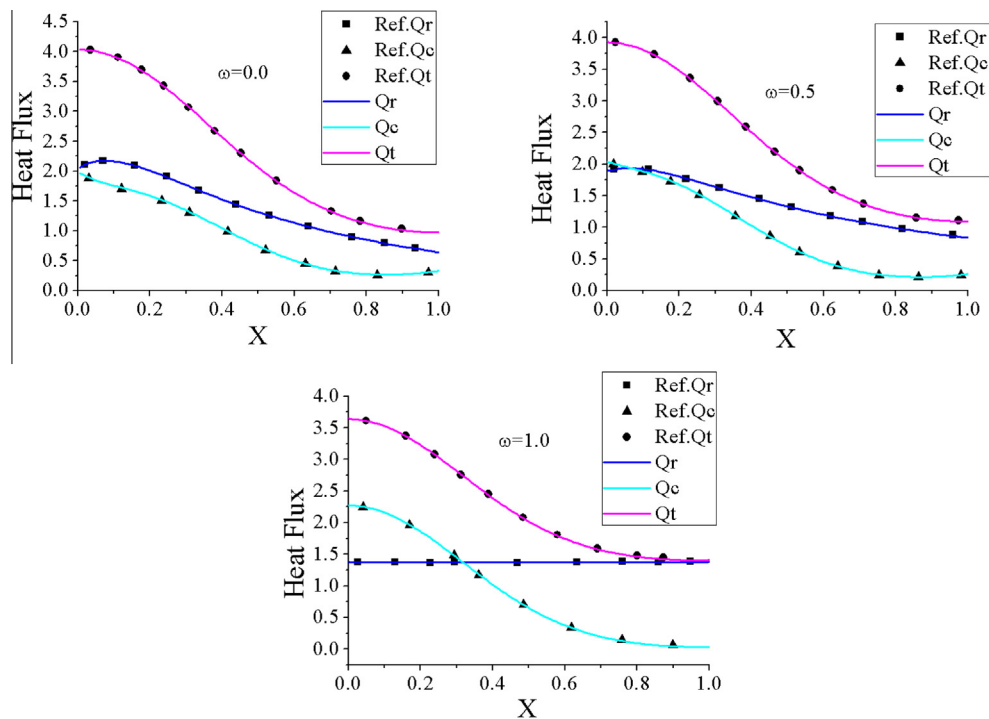


Fig. 4. Heat flux variations in the medium for different scattering albedo at $\zeta = 0.005$.

temperature does not change with time. The program run for a relative long time to satisfy this condition, although this can also be realized by stopping the simulation if results difference between two time steps is small enough. We adopt 16 angle control volumes in the zenith direction as it has small relative error.

We first give the results of case $\tau_L = 1.0$, $\omega = 0.5$, $T_E = 0.0$ and $N = 0.1$, which has been investigated by many other researchers. The conduction–radiation number N is defined as, $N = \kappa_a k / 4\sigma T_w^3$, where κ_a is the absorption coefficient, k is the thermal conductivity of the medium.

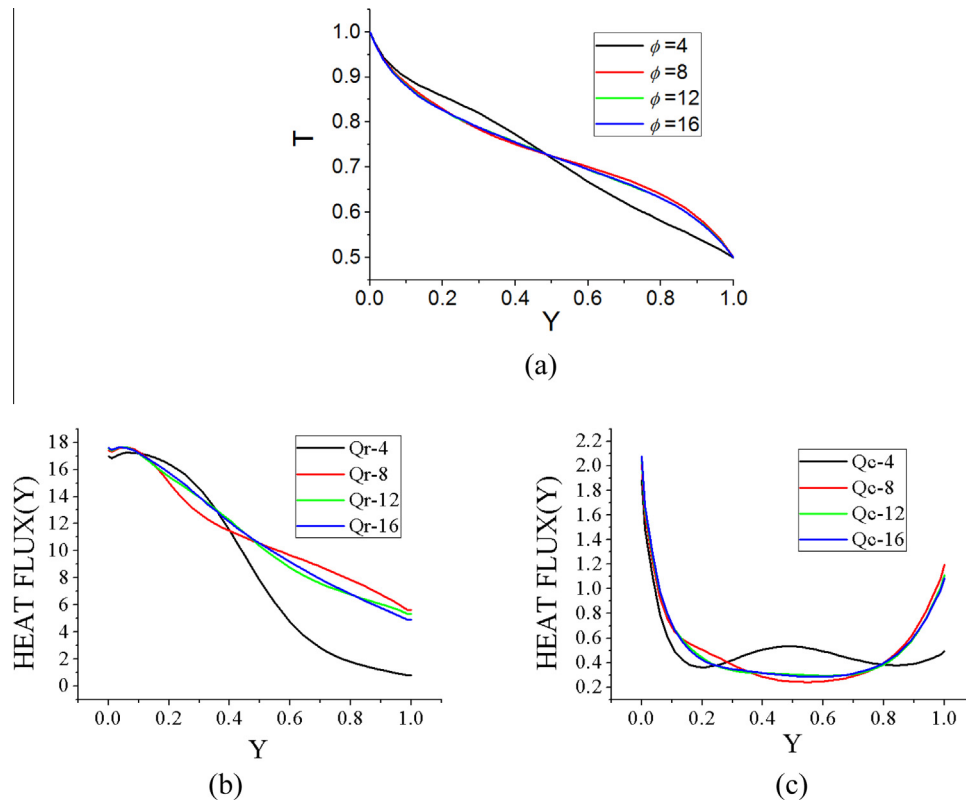


Fig. 5. Effect of azimuthal angle discretization on the temperature and the heat flux along the vertical central line.

Table 1 shows the dimensionless temperature at three points for $\zeta = 0.05$. It can be seen that good agreement with existing literature is achieved. This indicates that the method proposed in this work is capable of simulating the 1D combined radiation conduction problem.

Fig. 2 shows the temperature evolutions for different scattering albedos for $N = 0.01$, in which condition radiation heat transfer dominates. It can be seen that the temperature profile is nonlinear due to complex interactions between radiation and conduction and this indicates that LBM–FVM coupling has the ability to capture the nonlinear interaction. Fig. 2b and c shows the temperature distributions for $\omega = 0.5$, $\omega = 0.9$, respectively. It is clearly seen that the temperature profile decreases with increasing the scattering albedo, which decrease the radiation effect. But it is still quite different from the pure conduction case. This trend is the same in other references and proves the capability of our method.

Fig. 3 shows the transient dimensionless temperature evolutions in the 1D planar medium for different radiation–conduction number for $\omega = 0$. Fig. 3a shows the temperature distribution for

$N = 0.01$, with radiation heat transfer dominating. Fig. 3c represents the result for $N = 1.0$. It is obvious that the temperature distribution is almost linear at steady state and the temperature in the medium is also lower than that with smaller radiation–conduction number. At the initial stage, such as $\zeta = 0.001$, 0.005 , 0.015 , the temperature changes little, which is far different from the case of $N = 0.01$. This indicates that for cases dominated by radiation, the temperature in the whole medium rises quickly leading to decreased time to reach the steady state.

Except for the temperature profile, we also perform simulations to test the validity of this method for the heat flux calculation. Fig. 4 shows the heat flux profile at $\zeta = 0.005$ along the planar medium for the case of $N = 0.1$, $T_E = 0.1$ for different scattering albedo. All the results are in good agreement with Talukdar and Mishra [16], proving the ability of calculating heat flux with this method.

3.2. 2D cases

For a two-dimensional study, we consider a 2D square initially at temperature T_0 . When $t \geq 0$, the temperature of the south wall is raised to T_S and fixed. In the presence of radiation heat transfer, the heat transfer process will be different from that of pure conduction. This difference is mainly determined by the radiation–conduction number. The temperature of the interior medium will keep rising until steady state is reached. All the physical properties are assumed to be constant and the optical thickness along two coordinates are both chosen as 1.0 in this paper. We will first show mesh-independent tests and validations for the multi-dimensional study and then present and analyze the transient results.

For the spatial discretization, we use 41 control volumes in both directions, since results did not change when further refining the mesh. For angular discretization, the effect of number of azimuth angles on the temperature profile and heat flux profile are

Table 2

Dimensionless temperature at steady-state for $\tau_L = 1.0$, $\omega = 0.0$, $T_0 = 0.5$ and three different radiation conduction number.

N	Centerline T/T_S at Y	Wu and Ou [32]	Yuen and Takara [33]	Mondal and Mishra [15]	LBM–FVM present
1.0	0.3	0.733	0.737	0.738	0.734
	0.5	0.630	0.630	0.631	0.630
	0.7	0.560	0.560	0.565	0.565
0.1	0.3	0.760	0.763	0.761	0.757
	0.5	0.663	0.661	0.667	0.664
	0.7	0.590	0.589	0.598	0.596
0.01	0.3	0.791	0.807	0.777	0.786
	0.5	0.725	0.726	0.722	0.725
	0.7	0.663	0.653	0.672	0.668

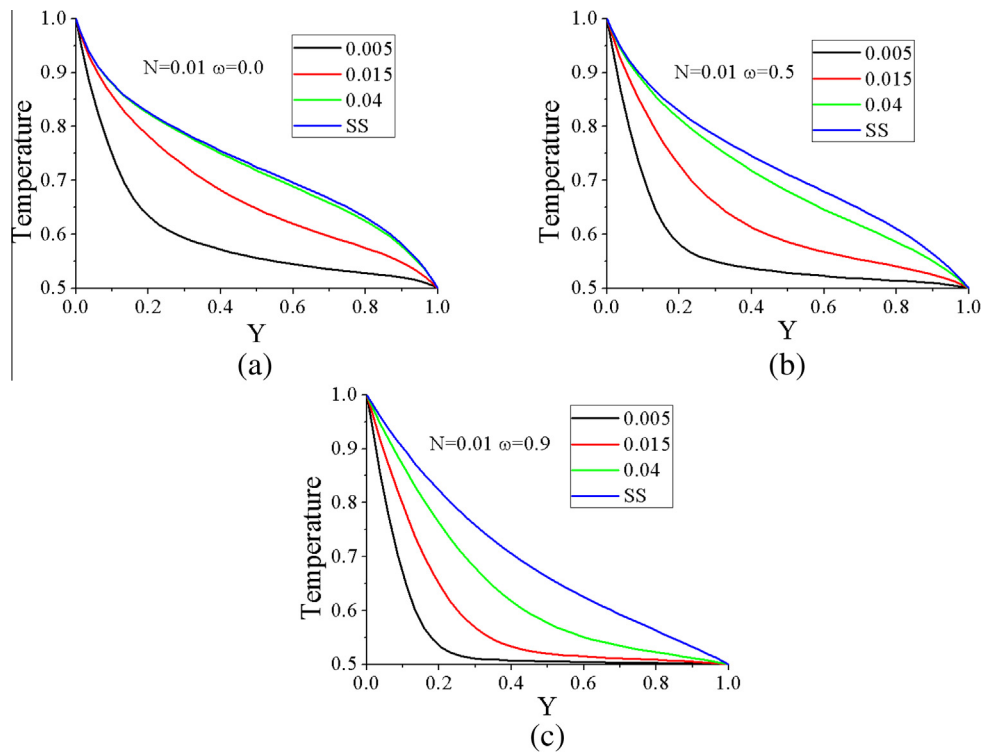


Fig. 6. Transient dimensionless temperature distributions for the 2D square for different scattering albedos (a) $\omega = 0.0$, (b) $\omega = 0.5$, and (c) $\omega = 0.9$.

investigated. Fig. 5a shows the temperature profile for four azimuth discretizations, i.e., $\phi = 4, 8, 12, 16$ (the number, not the angle value). Result of $\phi = 4$ is not good and hence not appropriate for this problem. There is little difference between $\phi = 8, \phi = 12$ and $\phi = 16$. So in terms of temperature, $\phi = 8$ may be adequate. While a further study of heat flux profiles indicates that this is not enough. As shown in Fig. 5b and c, for radiative and conductive heat flux

along the central vertical line, there is a big deviation between $\phi = 8$ and $\phi = 12$ or $\phi = 16$. When calculating heat flux is important, azimuth angular discretization should be larger than 8, like 12. This is not pointed out by Mishra and Roy [14].

Table 2 shows the temperature at three points along the vertical centerline at steady-state for $\tau_L = 1.0$, $\omega = 0.0$, $T_0 = 0.5$ and for three different radiation conduction number. These cases have

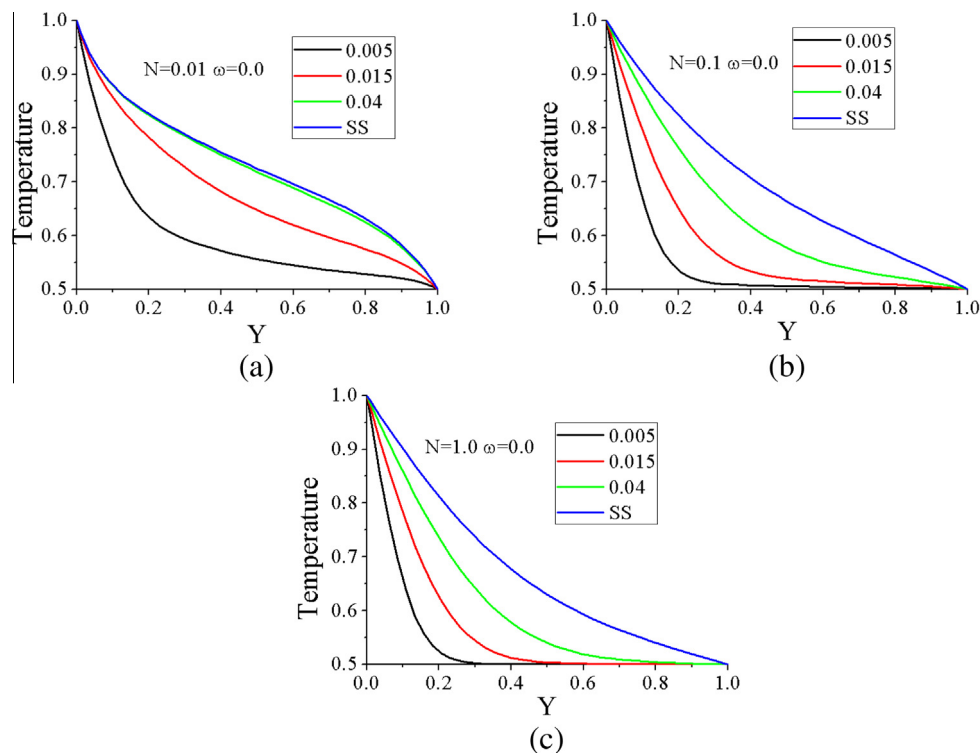


Fig. 7. Transient dimensionless temperature distribution along the planar medium for different conduction radiation parameters (a) $N = 0.01$, (b) $N = 0.1$, (c) $N = 1.0$.

been investigated by many researchers. As can be seen, good agreements are obtained with the existing literature and it proves the capability of this method to solve 2D combined radiation conduction heat transfer problems.

Fig. 6 shows the temperature evolutions along the vertical central line for the 2D square for different scattering albedos for $N = 0.01$, $T_0 = 0.5$. The temperature profile is similar to the 1D case. The temperature of this 2D medium is lower than that for the 1D case, which is due to heat loss through the west and east wall. The temperature profile decreases with the increase of the scattering albedo and its gradient near the north wall is larger for the smaller scattering albedo. From Fig. 6a and c, increasing scattering albedo blocks the radiation heat transfer and thus increases the time to get to the steady state.

Fig. 7 shows the temperature evolutions along the vertical central line for different radiation–conduction number for $\omega = 0.0$. As expected, the temperature of the medium decreases with increasing the radiation–conduction number. As seen from Fig. 7c, the temperature at $Y > 0.5$ changes little at $\zeta = 0.015$ for $N = 1.0$, while it is larger than 0.6 for the majority of the medium for the case of $N = 0.01$. For the temperature distribution of steady state, a larger temperature gradient is observed near the north wall for smaller radiation conduction number, which is similar to the 1D case.

4. Conclusions

This paper combines the lattice Boltzmann method and the finite volume method for combined radiation and conduction problems. The LBM is used to solve the energy equation and the FVM for the radiation transfer equation. The new proposed half-way boundary treatment of the thermal LBM is used in this work to avoid the interpolation processing between these two solvers previously used for combined heat transfer problems. For the Dirichlet boundary condition, this method is simpler than the original one. The 1D and 2D coupled conduction and radiation problems were used to test the capability of this method. The effects of scattering albedo and radiation–conduction number on the heat transfer are presented and analyzed. Results are in good agreement with the existing literature and thus this method is validated.

By overlapping the solver nodes of the LBM and the FVM, we avoid the interpolation operations between these two solvers, making it easier for their coupling. Although this paper investigated only the combined radiation and conduction problems, application to more complex systems (such as convection) is the same and the simplicity remains.

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